X-12
2/10/2026

END TO END COMMUNICATION AT TRANSPORT LAYER

AIM:-

To implement TCP/UDP echo client-server communication and a chat program using socket programming.

TCP

```
import socket
import threading

def start_server():
 server=socket.socket(socket.Af_INET, sockat.SOCK_STREAM
 .server.bind(('127.0.0.1', 5000))
 server.listen(1)
 Print("[SERVER] Listening on 127.0.0.1:5000")

 Conn, addr = server.accept()
 Print(f"[SERVER] Connected by {addr}")
 while True:
  data = conn.recv(1024).decode()
  if not data or data.lower()== 'quit':
   print("[SERVER] Connection closed by client.")
   break
  print(f"[SERVER] Received: {data}")
  conn.send(f"Server received: {data}".encode())
Conn.close()
server.close()
```

```
def start_client():
    client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    client.connect(('127.0.0.1', 5000))
    print("[CLIENT] Connected to server")

    while True:
        msg = input("Enter message (or 'quit' to exit): ")
        client.send(msg.encode())
        if msg.lower() == 'quit':
            break
        response = client.recv(1024).decode()
        print(f"[SERVER RESPONSE] {response}")
    client.close()

#---------MAIN--------
if __name__ == "__main__":
    server_thread = threading.Thread(target=start_server, daemon=True)
    server_thread.start()
    start_client()
```

OUTPUT:-

```
========= RESTART: C:/Users/Ikala/Downloads/tcp.py =========
[SERVER] Listening on 127.0.0.1:5000 [CLIENT] Connected to Server
[SERVER] Connected by ('127.0.0.1', 53458) Enter message (or 'quit' to exit):
hello
[SERVER] Received: hello
[SERVER RESPONSE] Server received: hello
Enter message (or 'quit' to exit):
```

```
========== RESTART: C:/Users/Hkala/Downloads/tcp.py ==========
[SERVER] Listening on 127.0.0.1:5000 [CLIENT] Connected to server
[SERVER] Connected by ('127.0.0.1', 60328) Enter message (or 'quit' to exit):
HI
[SERVER] Received: HI
[SERVER RESPONSE] Server received: HI
Enter message (or 'quit' to exit): quit
[SERVER] Connection closed by client.
```

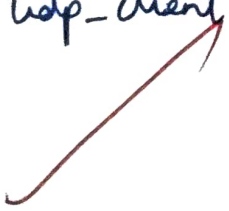
UDP

```
import socket
import threading

def udp_server():
    s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    s.bind(('127.0.0.1', 12000))
    print("[SERVER] Running on 127.0.0.1:12000")
    while True:
        data, addr = s.recvfrom(1024)
        msg = data.decode()
        if msg.lower() == 'quit':
            print(f"[SERVER] Client {addr} ended the connection.")
            break
        print(f"[SERVER] Received from {addr}: {msg}")
        s.sendto(data, addr)
    s.close()
    print("[SERVER] Stopped.")
```

```
#------ CLIENT--------
def udp_client():
    c=socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    server=('127.0.0.1', 12000)
    while True:
        msg=input("Enter message (or 'quit' to stop): ")
        c.sendto(msg.encode(), server)
        if msg.lower()=='quit':
            print("[CLIENT] Closing connection...")
            break
        data, _=c.recvfrom(1024)
        print("Echo from Server:", data.decode())
c.close()
print("[CLIENT] Stopped.")

#------ MAIN-----
if __name__=="__main__":
    threading.Thread(target=udp_server, daemon=True).start()
    udp_client()
```



OUTPUT:-

Enter message (or 'quit' to stop): [SERVER] Running on 127.0.0.1:12000 hi

[SERVER] Received from ('127.0.0.1', 49670): hi

Echo from Server: hi

Enter message (or 'quit' to stop): hello

[SERVER] Received from ('127.0.0.1', 49670): hello

Echo from Server: hello

Enter message (or 'quit' to stop): how are you

[SERVER] Received from ('127.0.0.1', 49670): how are you

Echo from Server: how are you

Enter message (or 'quit' to stop): quit

[CLIENT] Closing connection.... [SERVER] Client ('127.0.0.1', 49670) ended the connection.

[CLIENT] Stopped. [SERVER] Stopped.

RESULT:-

Thus, TCP/UDP client-server communication and message exchange were implemented and executed successfully.